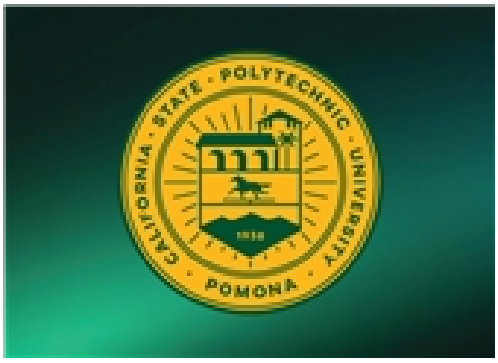
Project Name: ControlShield



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DESCRIPTION

ECE4311 – Network Forensics: Development of a PLC Guardian HMI for the M221 PLC on an HMIS5T panel, focusing on cybersecurity and network forensics for industrial control systems.

BILL OF MATERIAL

Controller



Reference	TM221CE24T
Description	TM221CE24T (screw) 14 digital inputs, 10 source transistor outputs (0,5 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 24 Vdc power supply controller with removable terminal blocks.
Power supplied to the IO bus	5V: 520 mA / 24V: 200 mA

HARDWARE CONFIGURATION

MyPLC - TM221CE24T

Digital Inputs

Used	Address	Filtering	Latch	Run/Stop	Events	Priority	Subroutine
	%I0.0	3 ms			Not Used		
	%I0.1	3 ms			Not Used		
	%I0.2	3 ms			Not Used		
	%I0.3	3 ms			Not Used		
	%I0.4	3 ms			Not Used		
	%I0.5	3 ms			Not Used		
	%I0.6	3 ms			Not Used		
	%I0.7	3 ms			Not Used		
	%I0.8	3 ms			Not Used		
	%I0.9	3 ms			Not Used		
	%I0.10	3 ms			Not Used		
	%I0.11	3 ms			Not Used		
	%I0.12	3 ms			Not Used		
	%I0.13	3 ms			Not Used		

Digital Outputs

Used	Address	Status Alarm	Fallback value	Used by
X	%Q0.0		0	User logic
X	%Q0.1		0	User logic
X	%Q0.2		0	User logic
	%Q0.3		0	
	%Q0.4		0	
	%Q0.5		0	
	%Q0.6		0	
	%Q0.7		0	
	%Q0.8		0	
	%Q0.9		0	

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW0.0	0 - 10 V	Normal	0-1000	0	
	%IW0.1	0 - 10 V	Normal	0-1000	0	

Fast Counters

Used	Address	Input	Configured	Preset	Double Word
	%FC0	%I0.2	NotUsed	0	
	%FC1	%I0.3	NotUsed	0	
	%FC2	%I0.4	NotUsed	0	
	%FC3	%I0.5	NotUsed	0	

High Speed Counters

Used	Address	Type
	%HSC0	Not Configured
	%HSC1	Not Configured
	%HSC2	Not Configured
	%HSC3	Not Configured

Pulse Generators

Configured	Address	Type
	%PLS0/%PWM0/%PTO0/%FREQGEN0	Not Configured
	%PLS1/%PWM1/%PTO1/%FREQGEN1	Not Configured

ETH1

Device name:	M221
IP Mode:	Fixed
IP address:	192.168.8.11
Subnet mask:	255.255.255.0
Gateway address:	0.0.0.0
Transfer Rate:	Auto
Security Parameters:	Programming protocol enabled
	Auto discovery protocol enabled
	Modbus server enabled
	EtherNet/IP protocol enabled

Modbus TCP

Unit ID:	247
Output registers (%IWM):	10
Input registers (%QWM):	10

SL1 (Serial line)

Physical Settings

Device:	None
Baud rate:	19200
Parity:	Even
Data bits:	8
Stop bits:	1
Physical medium:	RS-485
Polarization:	No

Protocol Settings

Protocol:	Modbus
Response timeout (× 100 ms):	10
Time between frames (ms):	10
Transmission mode:	RTU
Addressing:	Slave
Address:	1

SOFTWARE CONFIGURATION

Constant Words

KW

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment

KD

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment

KF

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment

Network Objects

Input Assembly (Ethernet/Ip)

Used	Address	Symbol	Fallback value	Comment

Output Assembly (Ethernet/Ip)

Used	Address	Symbol	Comment

Input Registers (Modbus Tcp)

Used	Address	Symbol	Fallback value	Comment

Output Registers (Modbus Tcp)

Used	Address	Symbol	Comment

Digital inputs (IOScanner)

Used	Address	Channel	Symbol	Comment

Digital outputs (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment

Input registers (IOScanner)

Used	Address	Channel	Symbol	Comment

Output registers (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment

Software Objects

Timers

Allocation: Automatic

Allocated: 2

Used	Address	Symbol	Type	Retentive	Dynamic Preset	Time Base	Preset	Comment
X	%TM0	TICK	TON			1 ms	100	
X	%TM1	TOCK	TOF			1 ms	100	

Counters

Allocation: Automatic

Allocated: 1

Used	Address	Symbol	Preset	Comment
X	%C0	TANK_N	80	

LIFO/FIFO Registers

Allocation: Automatic

Allocated: 0

Drums

Allocation: Automatic

Allocated: 0

Shift Bit Registers

Allocation: Automatic

Allocated: 0

Step Counters

Allocation: Automatic

Allocated: 0

Schedule Blocks

Allocation: Automatic

Allocated: 0

RTC



CPP

PID

Used	PID	Symbol	Type	Comment

Grafcet Steps

Allocation: Automatic

Allocated: 0

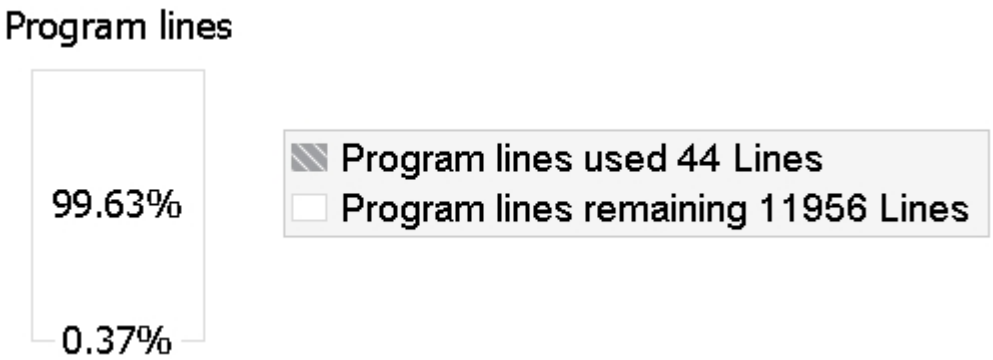
PROGRAM

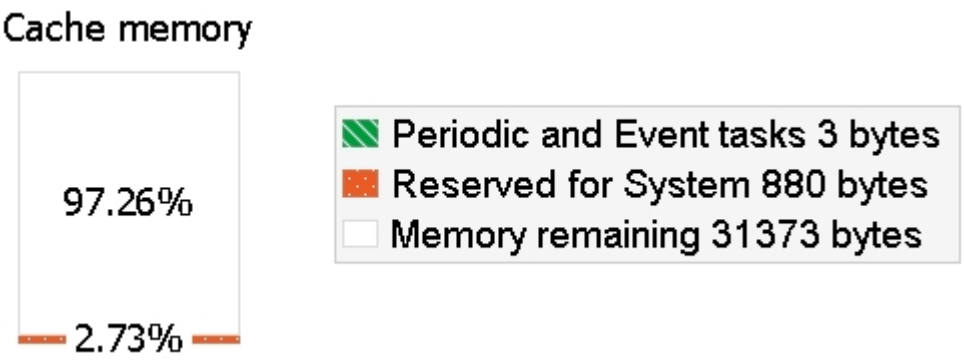
Behavior

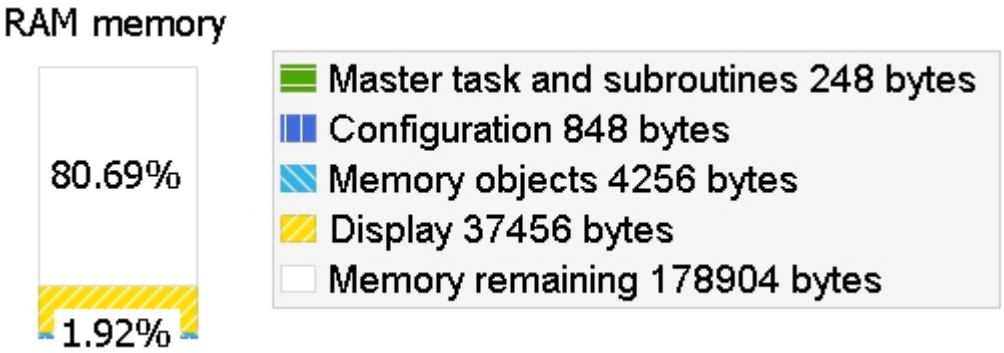
Functional level:	Level 14.0
Starting mode:	Start In Previous State
Watchdog:	250 ms
Fallback behavior:	Fallback value
String end character:	CR (Carriage Return)

Memory consumption

Last compilation: 03/11/2026 00:58:55







Application Architecture

Master Task

Scan mode:	Normal
POU list:	1 - Main

Periodic Task

Period:	255 ms

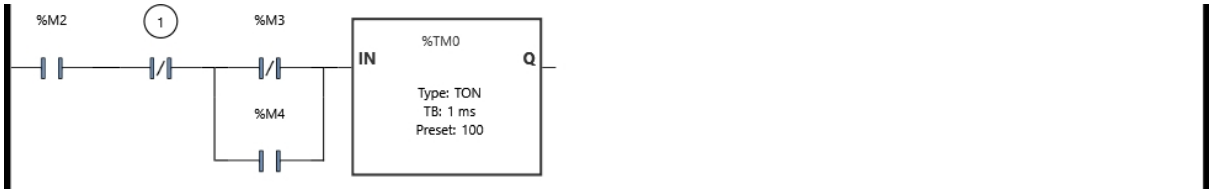
POU

Master Task

1 - Main

Master Task

Rung0 - Start/Stop



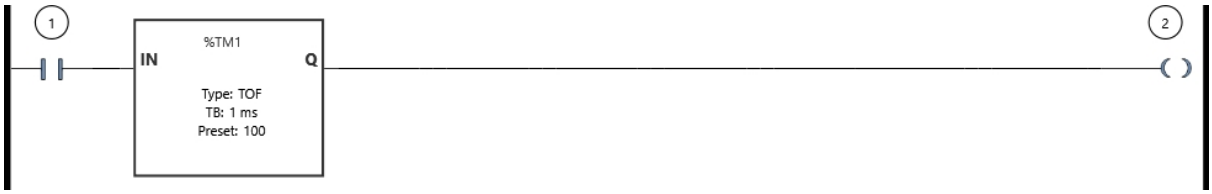
Legend:

1 %TM1.Q

Variables used:

%M2	STARTBUTTON
%M3	STOP
%M4	MAINT_BYPASS
%TM0	TICK
%TM1.Q	TOCK.Q

Rung1 - Simulated Encoder



Legend:

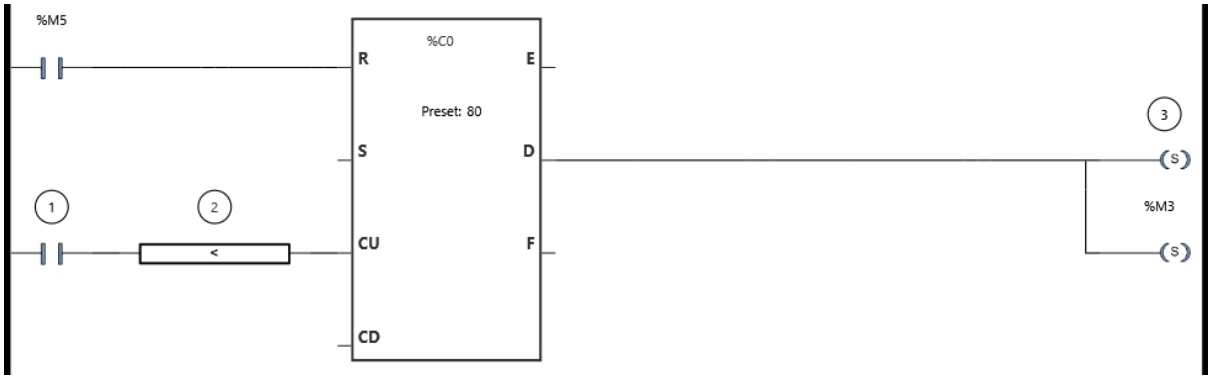
1 %TM0.Q

2 %Q0.0

Variables used:

%Q0.0	OUTPUT_1
%TM0.Q	TICK.Q
%TM1	TOCK

Rung2 - Tank Logic



Legend:

1 %Q0.0

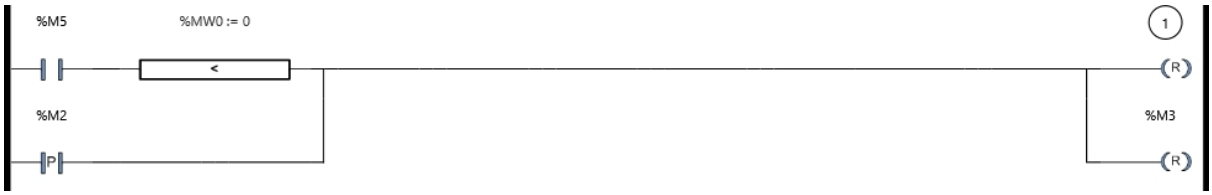
2 %MW0 := %C0.V

3 %Q0.1

Variables used:

%C0	TANK_N
%C0.V	TANK_N.V
%M3	STOP
%M5	RESET
%MW0	TANK_VALUE
%Q0.0	OUTPUT_1
%Q0.1	OUTPUT_2

Rung3 - Reset Logic



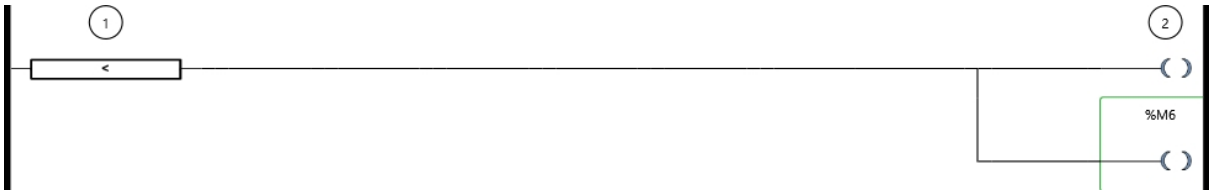
Legend:

1 %Q0.1

Variables used:

%M2	STARTBUTTON
%M3	STOP
%M5	RESET
%MW0	TANK_VALUE
%Q0.1	OUTPUT_2

Rung4 - Tank Overload Flag



Legend:

1 %C0.V > 100

2 %Q0.2

Variables used:

%C0.V	TANK_N.V
%M6	DIGI_OVERLOAD
%Q0.2	TANK_OVERLOAD

SYMBOLS

Used	Address	Symbol	Comment
X	%C0	TANK_N	
X	%M2	STARTBUTTON	
X	%M3	STOP	
X	%M4	MAINT_BYPASS	
X	%M5	RESET	
X	%M6	DIGI_OVERLOAD	
X	%MW0	TANK_VALUE	
X	%Q0.0	OUTPUT_1	
X	%Q0.1	OUTPUT_2	
X	%Q0.2	TANK_OVERLOAD	
X	%TM0	TICK	
X	%TM1	TOCK	

CROSS-REFERENCE TABLE

Address	Object	Rung	Code
%C0...	1 - Main	Rung2 - Tank Logic	%C0
%C0.V.	1 - Main	Rung2 - Tank Logic	--[<]-- %MW0 := %C0.V
		Rung4 - Tank Overload Flag	--[<]-- %C0.V > 100
%M2...	1 - Main	Rung0 - Start/Stop	--| |--
		Rung3 - Reset Logic	--|P|--
%M3...	1 - Main	Rung0 - Start/Stop	--|/|--
		Rung2 - Tank Logic	--(S)--
		Rung3 - Reset Logic	--(R)--
%M4...	1 - Main	Rung0 - Start/Stop	--| |--
%M5...	1 - Main	Rung2 - Tank Logic	--| |--
		Rung3 - Reset Logic	--| |--
%M6...	1 - Main	Rung4 - Tank Overload Flag	--( )--
%MW0..	1 - Main	Rung2 - Tank Logic	--[<]-- %MW0 := %C0.V
		Rung3 - Reset Logic	--[<]-- %MW0 := 0
%Q0.0.	1 - Main	Rung1 - Simulated Encoder	--( )--
		Rung2 - Tank Logic	--| |--
%Q0.1.	1 - Main	Rung2 - Tank Logic	--(S)--
		Rung3 - Reset Logic	--(R)--
%Q0.2.	1 - Main	Rung4 - Tank Overload Flag	--( )--
%TM0..	1 - Main	Rung0 - Start/Stop	%TM0
%TM0.Q	1 - Main	Rung1 - Simulated Encoder	--| |--
%TM1..	1 - Main	Rung1 - Simulated Encoder	%TM1
%TM1.Q	1 - Main	Rung0 - Start/Stop	--|/|--